

Zomato Delivery ETA

10-hour sprint · LightGBM + CatBoost blend · MAE 3.04 min

Team Hmmmmmmmmmm · Suchit SM · Rayyan Shaikh · Ranadeep M

github.com/TheClazer/Zomato-delivery-estimation

Final commit: [HASH] — tag: final

Predict Time_taken (min) for each delivery

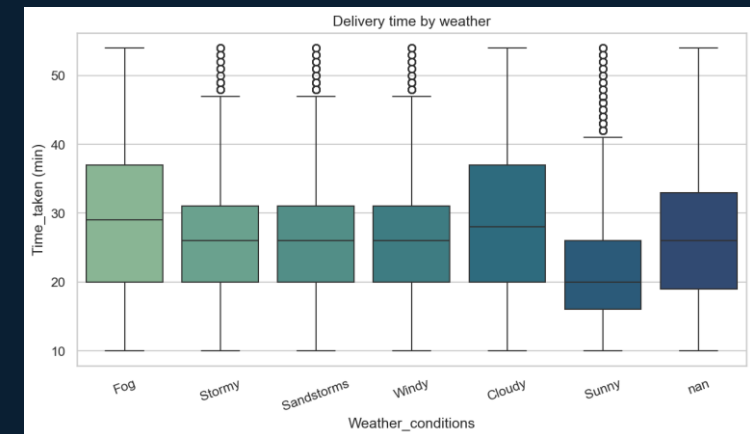
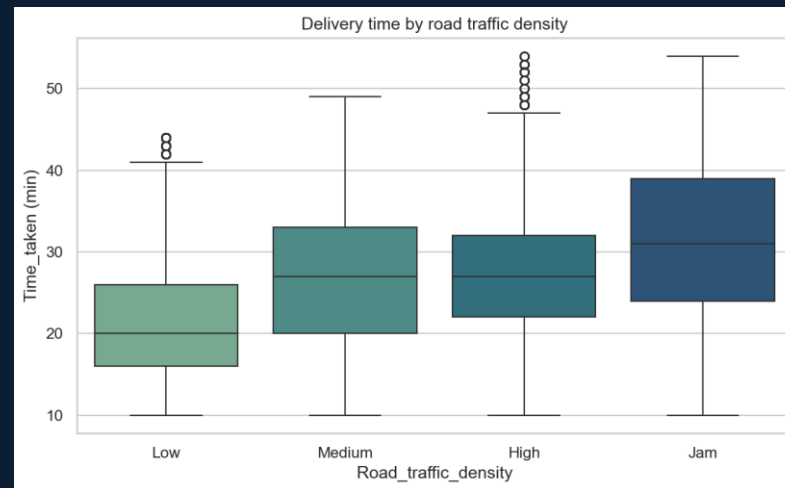
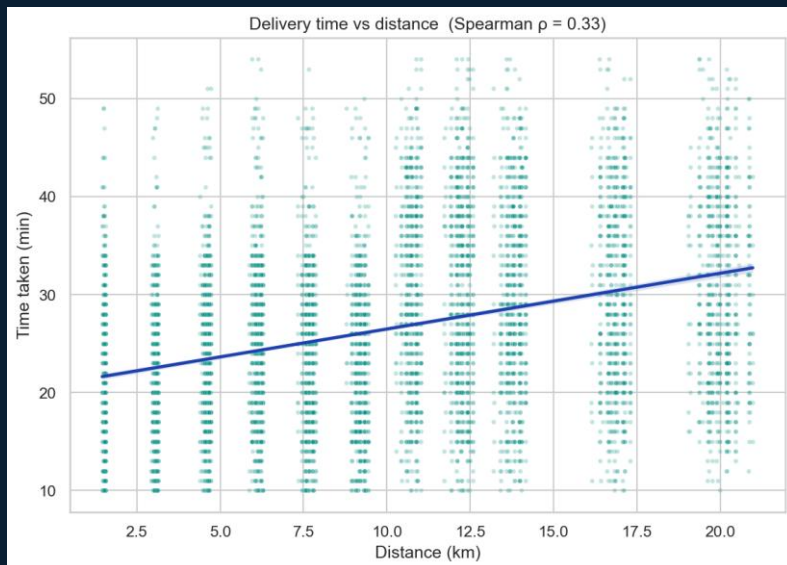
- Tabular regression: order + rider + restaurant + conditions → minutes.
- Metric: Mean Absolute Error (lower is better).
- Naïve baseline (predict the mean): $MAE \approx 7.6$ min → we land at 3.04.

Pipeline

- raw CSV → clean (src/data.py) → features (src/features.py)
- → KFold(5, seed=42) CV → LightGBM Optuna tune (60 trials, 5-min cap)
- → + CatBoost run on same split → weighted blend → freeze final.
- Reproducibility: every seed = 42, every train.parquet, model file, OOF saved.

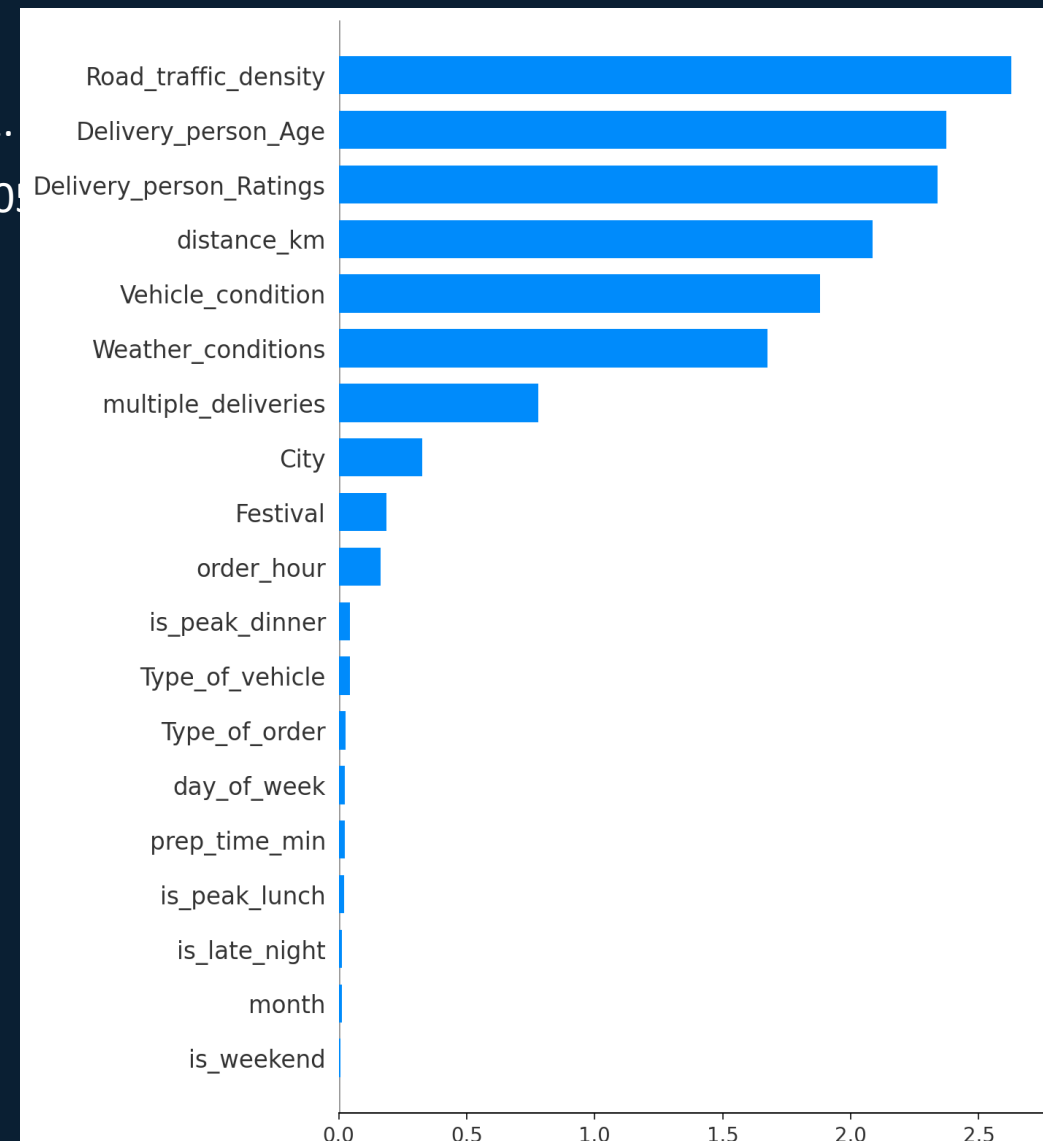
What our EDA found

- Distance $\rho = 0.33$ — clean primary signal.
- Traffic dominates: ANOVA $F=3390.2$, Low→Jam adds many minutes; verified significant at $p=0.00e+00$.
- Bad weather adds ≈ 6.0 min vs Sunny; festival days $\approx +19.5$ min.



Model and why

- LightGBM with native categorical handling (no one-hot bloat).
- Optuna 60-trial TPE search; 3-fold inner CV; final 5-fold KFold seed=42.
- Tuned LGBM MAE = 3.053 ± 0.018 (per-fold [3.03, 3.07, 3.04, 3.08, 3.05]).



Feature importance — SHAP bar

Post-pivot features (15:00 drop)

Auto-detect join key, build ≤ 3 features, retrain with the v1 tuned params on the same KFold.

READY — fills automatically when data/pivot/ receives files

- Pipeline staged in notebooks/_run_pivot.py — single command at 15:00.
- Auto-tries 6 join keys (Delivery_person_ID, Order_ID, ID, Restaurant_ID, City, Order_Date).
- Drops any column with $>80\%$ null after merge (noise filter).
- Builds ≤ 3 features so this slide stays readable.
- Retrains tuned v1 params on the new X matrix; same KFold seed=42.
- Writes reports/delta.json with a 'ship_v2' boolean — auto-decides.
- If $\text{delta} \leq 0.05$ min we keep v1 (rulebook §6: data literacy $>$ vanity).

The delta

v1 vs v2 — same CV split, no cherry-picking.

3.053

v1 MAE (locked)

TBA

v2 MAE (15:00 drop)

TBA

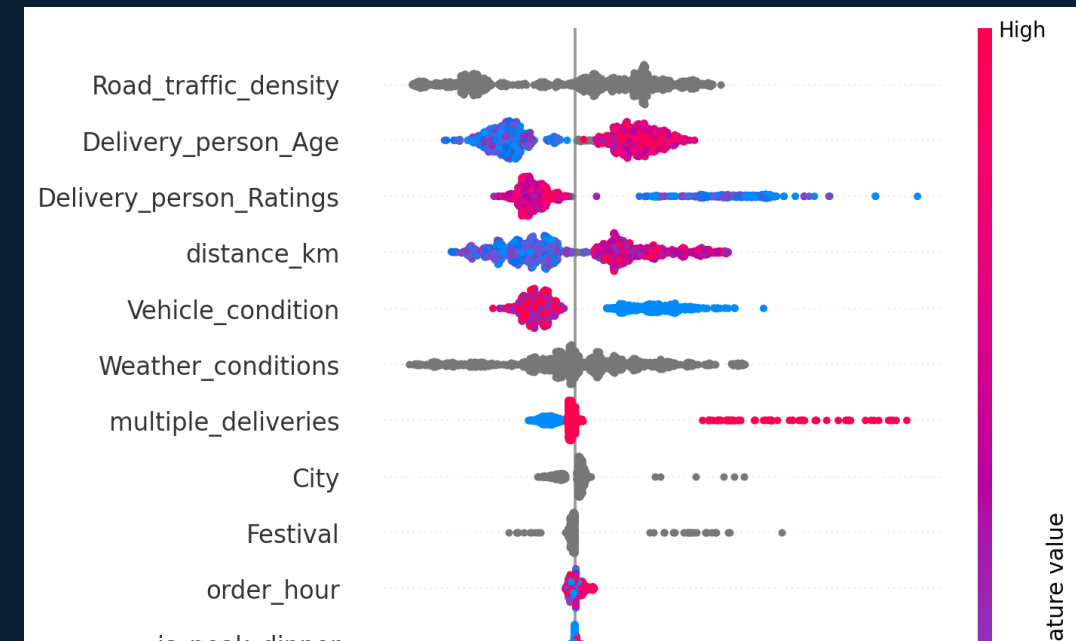
Δ MAE (computed live)

DECISION FRAMEWORK (locks at 17:00)

- If $\Delta > +0.05$ min $\rightarrow$ ship v2; this slide updates with the per-fold bar chart.
- If $\Delta \leq +0.05$ min $\rightarrow$ keep v1; slide 6 retitles to 'What the pivot data revealed'.
- Current v1 = 3.053 ± 0.018 ; blend = 3.043.
- Both branches are honest — rulebook §6 rewards correct verdicts over inflated claims.

Stability and interpretability

- 5-fold MAE std = 0.018 ($\approx 1\%$ of mean — model is stable).
- Top SHAP features: Road_traffic_density, Delivery_person_Age, Delivery_person_Ratings, distance_km, Vehicle_condition.
- No target leakage: prep_time is computed from order/pickup times only.
- Blend (0.55 LGBM + 0.45 CatBoost) further improves to 3.043 (-0.010).



Limitations and what's next

- No real-time traffic API; weather is categorical only.
- Coordinate noise contributes residual MAE — 431 rows had >30 km haversine (capped, not dropped).
- Next: quantile regression for ETA intervals + live traffic feed.
- Stacked ensemble (LGBM + CatBoost + linear meta-learner) on the existing OOF preds — code path already saved.

Thank you

Repo: github.com/TheClazer/Zomato-delivery-estimation

Tag: final

Q&A — distance, traffic, blend, pivot, anything.